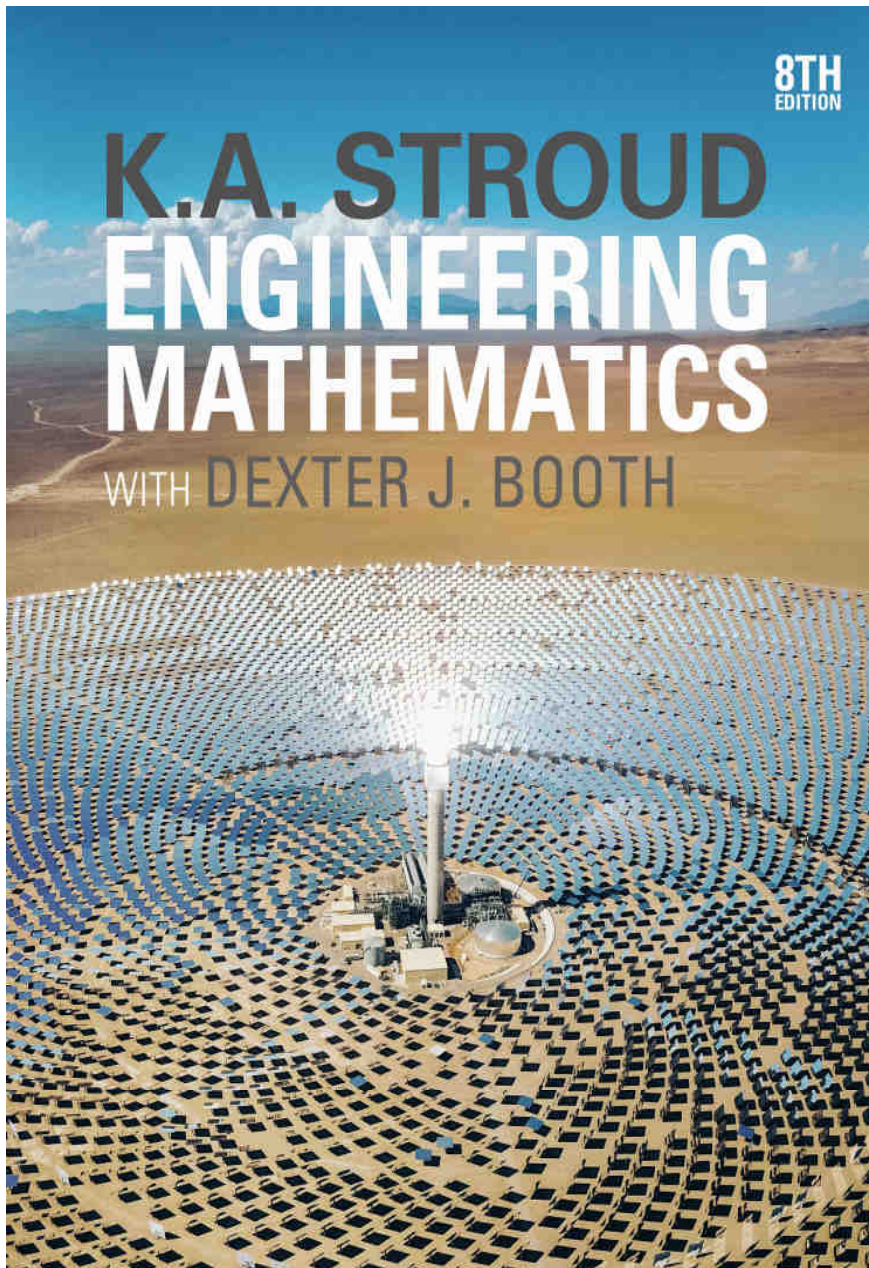




PROGRAMME 5

MATRICES

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Worked examples and exercises are in the text



Programme 5: **Matrices**

Matrices – definitions

Matrix notation

Equal matrices

Addition and subtraction of matrices

Multiplication of matrices

Transpose of a matrix

Special matrices

Determinant of a square matrix

Inverse of a square matrix

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Worked examples and exercises are in the text



Programme 5: Matrices

Learning outcomes

When you have completed this Programme you will be able to:

- ☐ Define a matrix
- ☐ Understand what is meant by the equality of two matrices
- ☐ Add and subtract two matrices
- ☐ Multiply a matrix by a scalar and multiply two matrices together
- ☐ Obtain the transpose of a matrix
- ☐ Recognize special types of matrix
- ☐ Obtain the determinant, cofactors and adjoint of a square matrix
- ☐ Obtain the inverse of a non-singular matrix

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Programme 5: Matrices

Matrices – definitions

A matrix is a set of real or complex numbers (called *elements*) arranged in rows and columns to form a rectangular array.

A matrix having m rows and n columns is called an $m \times n$ matrix.

For example:

$$\begin{pmatrix} 5 & 7 & 2 \\ 6 & 3 & 8 \end{pmatrix}$$

is a 2×3 matrix.

Programme 5: Matrices

Matrices – definitions

Row matrix

A row matrix consists of a single row. For example:

$$(4 \quad 3 \quad 7 \quad 2)$$

Column matrix

A column matrix consists of a single column. For example:

$$\begin{pmatrix} 6 \\ 3 \\ 8 \end{pmatrix}$$

Programme 5: Matrices

Matrices – definitions

So the matrix $\begin{pmatrix} 6 & 4 \\ 0 & 1 \\ 2 & 3 \end{pmatrix}$ is of order 3×2

and the matrix $\begin{pmatrix} 2 & 5 & 3 & 4 \\ 6 & 7 & 4 & 9 \end{pmatrix}$ is of order 2×4

$\begin{pmatrix} 5 \\ 2 \end{pmatrix}$ is a $column$ matrix of order 2×1

$(4 \ 0 \ 7 \ 3)$ is a row matrix of order 1×4

Programme 5: Matrices

Matrices – definitions

Double suffix notation

Each element of a matrix has its own address denoted by double suffices, the first indicating the row and the second indicating the column. For example, the elements of 3×4 matrix can be written as:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{pmatrix}$$

2nd row 3rd column

Programme 5: Matrices

Matrices – definitions

Therefore, in the matrix

$$\begin{pmatrix} 6 & -5 & 1 & -3 \\ 2 & -4 & 8 & 3 \\ 4 & -7 & -6 & 5 \\ -2 & 9 & 7 & -1 \end{pmatrix}$$

Handwritten annotations: A blue arrow points from the left to the second row. A blue arrow points from the bottom to the second column. A blue circle is drawn around the element 3 in the second row, fourth column, with a line pointing to the label a_{24} . A blue circle is drawn around the element -1 in the fourth row, fourth column, with a line pointing to the label 4. A blue circle is drawn around the element 9 in the fourth row, second column, with a line pointing to the label 2.

- the location of
- (a) the element 3 can be stated as a_{24}
 - (b) the element -1 can be stated as a_{44}
 - (c) the element 9 can be stated as a_{42}

Programme 5: Matrices

Matrices – definitions

Example 1.1

Given

$$C = \begin{pmatrix} 3 & 0 & 4 & x \\ \lambda & 1 & z & -1 \end{pmatrix}$$

state (a) c_{13} , (b) c_{21} , (c) c_{22} .

Programme 5: Matrices

Matrix notation

Where there is no ambiguity a matrix can be represented by a single general element in brackets or by a capital letter in bold type.

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{pmatrix} \text{ can be denoted by } (a_{ij}) \text{ or by } \mathbf{A}$$

So, if $\mathbf{B}$ represents a 2×3 matrix, write out the elements b_{ij} in the matrix, using the double suffix notation. This gives

$$\mathbf{B} = \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{pmatrix}$$

Example 1.2 Matrices and digital image processing

A digital image is an image that is represented by an array of numbers. Consider, as a very simple example, the mathematical representation of the symbol π . Very crudely we can picture the symbol as in Figure 1.1.

Figure 1.1

A simple digital representation of π .



We could represent this image of π by a 3×5 matrix such as

$$B = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$



$$B = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Note that we have divided the image into a number of areas, or fields, in this case 15 of them. We have then assigned a number to each field: 1 if the field is to be shaded, and 0 otherwise.

Images such as black and white photographs can be ‘digitised’ in the same way. The image is divided into $m \times n$ fields, where in practice m and n are large integers. Each field is assigned a number that represents the intensity of the image in that area. Often a 64-point grey scale is used so that a range of grey levels can be represented ranging from white at one extreme to black at the other. A digital image is then represented by an $m \times n$ matrix, B say, whose elements are numbers in the range 1–64. In image processing the matrix B is often referred to as $b[m, n]$.

Programme 5: Matrices

Equal matrices

Two matrices are equal if corresponding elements throughout are equal.

$\mathbf{A} = \mathbf{B}$ that is $(a_{ij}) = (b_{ij})$ if $a_{ij} = b_{ij}$ for all values of i and j

So, if
$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix} = \begin{pmatrix} 5 & -7 & 3 \\ 1 & 2 & 6 \\ 0 & 4 & 8 \end{pmatrix}$$
 then $d = \dots 1 \dots$; $b = \dots -7 \dots$; $a - k = \dots 5 - 8 = -3 \dots$

Programme 5: Matrices

Addition and subtraction of matrices

To be added or subtracted, two matrices must be of the same order.

Two matrices are added (or subtracted) by adding (or subtracting) corresponding elements. For example:

$$\begin{pmatrix} 4 & 2 & 3 \\ 5 & 7 & 6 \end{pmatrix} + \begin{pmatrix} 1 & 8 & 9 \\ 3 & 5 & 4 \end{pmatrix} = \begin{pmatrix} 4+1 & 2+8 & 3+9 \\ 5+3 & 7+5 & 6+4 \end{pmatrix} \\ = \begin{pmatrix} 5 & 10 & 12 \\ 8 & 12 & 10 \end{pmatrix}$$

So, (a) $\begin{pmatrix} 6 & 5 & 4 & 1 \\ 2 & 3 & -7 & 8 \end{pmatrix} + \begin{pmatrix} 1 & 4 & 2 & 3 \\ 6 & -1 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 7 & 9 & 6 & 4 \\ 8 & 2 & -7 & 13 \end{pmatrix}$

(b) $\begin{pmatrix} 8 & 3 & 6 \\ 5 & 2 & 7 \\ 1 & 0 & 4 \end{pmatrix} - \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} 7 & 1 & 3 \\ 1 & -3 & 1 \\ -6 & -8 & -5 \end{pmatrix}$

Programme 5: Matrices

Multiplication of matrices

1. *Scalar multiplication*
2. *Multiplication of two matrices*

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Programme 5: Matrices

Multiplication of matrices

Scalar multiplication

To multiply a matrix by a single number (a scalar), each individual element of the matrix is multiplied by that number. For example:

$$4 \times \begin{pmatrix} 3 & 2 & 5 \\ 6 & 1 & 7 \end{pmatrix} = \begin{pmatrix} 12 & 8 & 20 \\ 24 & 4 & 28 \end{pmatrix}$$

That is:

$$k(a_{ij}) = (ka_{ij})$$

Programme 5: Matrices

Multiplication of matrices

Multiplication of two matrices

Two matrices can only be multiplied when the number of columns in the first matrix equals the number of rows in the second matrix.

The ij th element of the product matrix is obtained by multiplying each element in the i th row of the first matrix by the corresponding element in the j th column of the second matrix and the element products added.

For example:

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Programme 5: Matrices

Multiplication of matrices

Multiplication of two matrices

$$\text{If } \mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} b_{11} \\ b_{21} \\ b_{31} \end{pmatrix}$$

$$\text{then } \mathbf{A} \cdot \mathbf{B} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \cdot \begin{pmatrix} b_{11} \\ b_{21} \\ b_{31} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} \end{pmatrix}$$

Programme 5: Matrices

Multiplication of matrices

Multiplication of two matrices

Example 1

$$\begin{pmatrix} 4 & 7 & 6 \\ 2 & 3 & 1 \end{pmatrix} \cdot \begin{pmatrix} 8 \\ 5 \\ 9 \end{pmatrix} = \begin{pmatrix} 4 \times 8 + 7 \times 5 + 6 \times 9 \\ 2 \times 8 + 3 \times 5 + 1 \times 9 \end{pmatrix} = \begin{pmatrix} 32 + 35 + 54 \\ 16 + 15 + 9 \end{pmatrix} = \begin{pmatrix} 121 \\ 40 \end{pmatrix}$$

Similarly $\begin{pmatrix} 2 & 3 & 5 & 1 \\ 4 & 6 & 0 & 7 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 4 \\ 2 \\ 9 \end{pmatrix} = \dots \dots \dots = \begin{pmatrix} 6 + 12 + 10 + 9 \\ 12 + 24 + 0 + 63 \end{pmatrix} = \begin{pmatrix} 37 \\ 99 \end{pmatrix}$

2×4 4×1

2×1

Programme 5: Matrices

Multiplication of matrices

Multiplication of two matrices

Example 2

$$\begin{aligned} \text{If } \mathbf{A} = (a_{ij}) &= \begin{pmatrix} 1 & 5 \\ 2 & 7 \\ 3 & 4 \end{pmatrix} \text{ and } \mathbf{B} = (b_{ij}) = \begin{pmatrix} 8 & 4 & 3 & 1 \\ 2 & 5 & 8 & 6 \end{pmatrix} \\ \text{then } \mathbf{A} \cdot \mathbf{B} &= \begin{pmatrix} 1 & 5 \\ 2 & 7 \\ 3 & 4 \end{pmatrix} \cdot \begin{pmatrix} 8 & 4 & 3 & 1 \\ 2 & 5 & 8 & 6 \end{pmatrix} \\ &= \begin{pmatrix} 18 & 29 & 43 & 31 \\ 30 & 43 & 62 & 44 \\ 32 & 32 & 41 & 27 \end{pmatrix} \end{aligned}$$

Programme 5: Matrices

Multiplication of matrices

Multiplication of two matrices

Note that multiplying a (3×2) matrix and a (2×4) matrix gives a product matrix of order (3×4)

$$\text{i.e. order } (3 \times \textcircled{2}) \times \text{order } (\textcircled{2} \times 4) \rightarrow \text{order } \underline{\underline{(3 \times 4)}}.$$

(same number 2)

In general then, the product of an $(l \times m)$ matrix and an $(m \times n)$ matrix has order $(l \times n)$.

It follows that a matrix can be squared only if it is itself a square matrix, i.e. the number of rows equals the number of columns.

Programme 5: Matrices

If **A** is an $(m \times n)$ matrix
and **B** is an $(n \times m)$ matrix } then products **A.B** and **B.A** are possible.

Example

$$\text{If } \mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 7 & 10 \\ 8 & 11 \\ 9 & 12 \end{pmatrix}$$

$$\text{then } \mathbf{A.B} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \cdot \begin{pmatrix} 7 & 10 \\ 8 & 11 \\ 9 & 12 \end{pmatrix} = \begin{pmatrix} 50 & 68 \\ 122 & 167 \end{pmatrix}$$

$$\text{and } \mathbf{B.A} = \begin{pmatrix} 7 & 10 \\ 8 & 11 \\ 9 & 12 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} 47 & 64 & 81 \\ 52 & 71 & 90 \\ 57 & 78 & 99 \end{pmatrix}$$

Note that, in matrix multiplication, $\mathbf{A.B} \neq \mathbf{B.A}$, i.e. multiplication is not commutative. The order of the factors is important!

In the product $\mathbf{A.B}$, **B** is *pre-multiplied* by **A**
and **A** is *post-multiplied* by **B**.

Exercise:

Programme 5: Matrices

Define the matrices

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix}_{2 \times 3}, \quad B = \begin{pmatrix} 4 & -2 & 1 \\ 2 & -4 & -2 \end{pmatrix}_{2 \times 3}, \quad C = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}_{2 \times 2},$$

$$D = \begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix}_{2 \times 2}, \quad E = \begin{pmatrix} 1 \\ 2 \end{pmatrix}_{2 \times 1}.$$

Compute if defined: $B - 2A$, $3C - E$, AC , CD , CB .

Handwritten dimension checks for AC , CD , and CB :

$$\begin{array}{c} \begin{array}{c|c} A & C \\ \hline 2 \times 3 & 2 \times 2 \end{array} \\ \times \end{array} \quad \begin{array}{c} \begin{array}{c|c} C & D \\ \hline 2 \times 2 & 2 \times 2 \end{array} \\ = \end{array} \quad \begin{array}{c} \begin{array}{c|c} C & B \\ \hline 2 \times 2 & 2 \times 3 \end{array} \\ = \end{array}$$

$$B - 2A = \begin{pmatrix} 4 & -2 & 1 \\ 2 & -4 & -2 \end{pmatrix} - \begin{pmatrix} 4 & 2 & -2 \\ 2 & -2 & 2 \end{pmatrix} = \begin{pmatrix} 0 & -4 & 3 \\ 0 & -2 & -4 \end{pmatrix}$$

$$CD = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} 11 & 10 \\ 10 & 11 \end{pmatrix}$$

$$CB = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}_{2 \times 2} \begin{pmatrix} 4 & -2 & 1 \\ 2 & -4 & -2 \end{pmatrix}_{2 \times 3} = \begin{pmatrix} 8 & -10 & -3 \\ 10 & -8 & 0 \end{pmatrix}_{2 \times 3}$$

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Programme 5: Matrices

Transpose of a matrix

If a new matrix is formed by interchanging rows and columns the new matrix is called the *transpose* of the original matrix. For example, if:

$$\mathbf{A} = \begin{pmatrix} 4 & 6 \\ 7 & 9 \\ 2 & 5 \end{pmatrix} \text{ then } \mathbf{A}^T = \begin{pmatrix} 4 & 7 & 2 \\ 6 & 9 & 5 \end{pmatrix}$$

Programme 5: Matrices

Transpose of a matrix

Therefore, given that

$$\mathbf{A} = \begin{pmatrix} 2 & 7 & 6 \\ 3 & 1 & 5 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 4 & 0 \\ 3 & 7 \\ 1 & 5 \end{pmatrix}$$

then $\mathbf{A.B} = \dots\dots\dots$ and $(\mathbf{A.B})^T = \dots\dots\dots$

$$\begin{pmatrix} 35 & 79 \\ 20 & 32 \end{pmatrix}$$

$$(\mathbf{A.B})^T = \begin{pmatrix} 35 & 20 \\ 79 & 32 \end{pmatrix}$$

Programme 5: Matrices

Transpose of a matrix

The following are useful and easy to prove facts:

$$\textcircled{1} \left(A^T\right)^T = A, \text{ and } \textcircled{2} (A + B)^T = A^T + B^T.$$

A less obvious fact is that the transpose of the product of matrices is equal to the product of the transposes with the order of multiplication reversed, i.e.,

$$\textcircled{3} (AB)^T = B^T A^T.$$

Programme 5: Matrices

Special matrices

Square matrix

Diagonal matrix

Triangular matrix

Unit matrix

Null matrix

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Programme 5: Matrices

Special matrices

Square matrix

A square matrix is of order $m \times m$.

A square matrix is *symmetric* if $a_{ij} = a_{ji}$. For example:

$$\begin{pmatrix} 1 & 2 & 5 \\ 2 & 8 & 9 \\ 5 & 9 & 4 \end{pmatrix} \quad \text{Then } \boxed{A^T = A}$$

A square matrix is *skew-symmetric* if $a_{ij} = -a_{ji}$. For example:

$$\begin{pmatrix} 0 & 2 & 5 \\ -2 & 0 & 9 \\ -5 & -9 & 0 \end{pmatrix} \quad \text{Then } \boxed{A^T = -A}$$

Programme 5: Matrices

Special matrices

Diagonal matrix

A diagonal matrix is a square matrix with all elements zero except those on the **leading diagonal**. For example:

$$\begin{pmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 7 \end{pmatrix}$$

Programme 5: Matrices

Special matrices

Upper and Lower triangular matrices

An upper or lower triangular matrix is a square matrix with all elements zero below or above the diagonal. For example, 3x3 upper and lower triangular matrices are given by:

$$U = \begin{pmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{pmatrix}, \quad L = \begin{pmatrix} a & 0 & 0 \\ b & d & 0 \\ c & e & f \end{pmatrix}.$$

$|B|$ is simply the product of the diagonal entries when B is triangular, i.e $|B| = b_{11}b_{22}b_{33} \dots b_{nn}$.

Programme 5: Matrices

Special matrices

Unit matrix — Identity matrix

A unit matrix is a diagonal matrix with all elements on the leading diagonal being equal to unity. For example:

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

3×3

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

2×2

The product of matrix $\mathbf{A}$ and the unit matrix is the matrix $\mathbf{A}$:

$$\mathbf{A} \cdot \mathbf{I} = \mathbf{A}$$

Programme 5: Matrices

Special matrices

Null matrix

A null matrix is one whose elements are all zero.

$$\mathbf{0} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Notice that

$$\mathbf{A} \cdot \mathbf{0} = \mathbf{0}$$

But that if $\mathbf{A} \cdot \mathbf{B} = \mathbf{0}$ we cannot deduce that $\mathbf{A} = \mathbf{0}$ or $\mathbf{B} = \mathbf{0}$

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Programme 5: Matrices

Special matrices

Null matrix

If $\mathbf{A} \cdot \mathbf{B} = \mathbf{0}$, we cannot say that therefore $\mathbf{A} = \mathbf{0}$ or $\mathbf{B} = \mathbf{0}$

$$\text{for if } \mathbf{A} = \begin{pmatrix} 2 & 1 & -3 \\ 6 & 3 & -9 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 1 & 9 \\ 4 & -6 \\ 2 & 4 \end{pmatrix}$$

$$\begin{aligned} \text{then } \mathbf{A} \cdot \mathbf{B} &= \begin{pmatrix} 2 & 1 & -3 \\ 6 & 3 & -9 \end{pmatrix} \cdot \begin{pmatrix} 1 & 9 \\ 4 & -6 \\ 2 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 2+4-6 & 18-6-12 \\ 6+12-18 & 54-18-36 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{aligned}$$

That is, $\mathbf{A} \cdot \mathbf{B} = \mathbf{0}$, but clearly $\mathbf{A} \neq \mathbf{0}$ and $\mathbf{B} \neq \mathbf{0}$.

Programme 5: Matrices

Determinant of a square matrix

Singular matrix

Cofactors

Adjoint of a square matrix

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Worked examples and exercises are in the text

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Programme 5: Matrices

Determinant of a square matrix

Singular matrix

Every square matrix has its associated determinant. For example, the determinant of

$$\begin{pmatrix} 5 & 2 & 1 \\ 0 & 6 & 3 \\ 8 & 4 & 7 \end{pmatrix} \text{ is } \begin{vmatrix} 5 & 2 & 1 \\ 0 & 6 & 3 \\ 8 & 4 & 7 \end{vmatrix} = 150$$

The determinant of a matrix is equal to the determinant of its transpose.

A matrix whose determinant is zero is called a *singular* matrix.

Programme 5: Matrices

Determinant of a square matrix

Cofactors

Each element a_{ij} of a square matrix has a *minor* which is the value of the determinant obtained from the matrix after eliminating the i th row and j th column to which the element is common.

The cofactor of element a_{ij} is then given as the minor of a_{ij} multiplied by

$$(-1)^{i+j}$$

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Programme 5: Matrices

Determinant of a square matrix

Adjoint of a square matrix

Let square matrix $\mathbf{C}$ be constructed from the square matrix $\mathbf{A}$ where the elements of $\mathbf{C}$ are the respective cofactors of the elements of $\mathbf{A}$ so that if:

$$\mathbf{A} = (a_{ij}) \text{ and } A_{ij} \text{ is the cofactor of } a_{ij} \text{ then } \mathbf{C} = (A_{ij})$$

Then the transpose of $\mathbf{C}$ is called the adjoint of $\mathbf{A}$, denoted by $\text{adj}\mathbf{A}$.

Programme 5: Matrices

Determinant of a square matrix

Adjoint of a square matrix

If we start afresh with $\mathbf{A} = \begin{pmatrix} 2 & 3 & 5 \\ 4 & 1 & 6 \\ 1 & 4 & 0 \end{pmatrix}$, its determinant

$\det \mathbf{A} = |\mathbf{A}| = \begin{vmatrix} 2 & 3 & 5 \\ 4 & 1 & 6 \\ 1 & 4 & 0 \end{vmatrix}$ from which we can form a new matrix $\mathbf{C}$ of the cofactors.

Programme 5: Matrices

Determinant of a square matrix

Adjoint of a square matrix

$$\mathbf{C} = \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix} \text{ where } \begin{array}{l} A_{11} \text{ is the cofactor of } a_{11} \\ A_{ij} \text{ is the cofactor of } a_{ij} \text{ etc.} \end{array}$$

$$A_{11} = + \begin{vmatrix} 1 & 6 \\ 4 & 0 \end{vmatrix} \\ = +(0 - 24) = -24$$

$$A_{12} = - \begin{vmatrix} 4 & 6 \\ 1 & 0 \end{vmatrix} \\ = -(0 - 6) = 6$$

$$A_{13} = + \begin{vmatrix} 4 & 1 \\ 1 & 4 \end{vmatrix} \\ = +(16 - 1) = 15$$

$$A_{21} = - \begin{vmatrix} 3 & 5 \\ 4 & 0 \end{vmatrix} \\ = -(0 - 20) = 20$$

$$A_{22} = + \begin{vmatrix} 2 & 5 \\ 1 & 0 \end{vmatrix} \\ = +(0 - 5) = -5$$

$$A_{23} = - \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} \\ = -(8 - 3) = -5$$

$$A_{31} = + \begin{vmatrix} 3 & 5 \\ 1 & 6 \end{vmatrix} \\ = +(18 - 5) = 13$$

$$A_{32} = - \begin{vmatrix} 2 & 5 \\ 4 & 6 \end{vmatrix} \\ = -(12 - 20) = 8$$

$$A_{33} = + \begin{vmatrix} 2 & 3 \\ 4 & 1 \end{vmatrix} = +(2 - 12) \\ = -10$$

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Programme 5: Matrices

Determinant of a square matrix

Adjoint of a square matrix

$$\therefore \text{The matrix of cofactors is } \mathbf{C} = \begin{pmatrix} -24 & 6 & 15 \\ 20 & -5 & -5 \\ 13 & 8 & -10 \end{pmatrix}$$
$$\text{and the transpose of } \mathbf{C}, \text{ i.e. } \mathbf{C}^T = \begin{pmatrix} -24 & 20 & 13 \\ 6 & -5 & 8 \\ 15 & -5 & -10 \end{pmatrix}$$

This is called the *adjoint* of the original matrix $\mathbf{A}$ and is written $\text{adj } \mathbf{A}$.

Therefore, to find the adjoint of a square matrix $\mathbf{A}$:

- (a) we form the matrix $\mathbf{C}$ of cofactors,
- (b) we write the transpose of $\mathbf{C}$, i.e. $\mathbf{C}^T$.

Programme 5: Matrices

Inverse of a square matrix

If each element of the adjoint of a square matrix $\mathbf{A}$ is divided by the determinant of $\mathbf{A}$ then the resulting matrix is called the inverse of $\mathbf{A}$, denoted by $\mathbf{A}^{-1}$.

$$\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}} (\text{adj} \mathbf{A})$$

Note: if $\det \mathbf{A} = 0$ then the inverse does not exist

Programme 5: Matrices

Inverse of a square matrix

For the matrix which we used in the last frame, $\mathbf{A} = \begin{pmatrix} 2 & 3 & 5 \\ 4 & 1 & 6 \\ 1 & 4 & 0 \end{pmatrix}$

$$\det \mathbf{A} = |\mathbf{A}| = \begin{vmatrix} 2 & 3 & 5 \\ 4 & 1 & 6 \\ 1 & 4 & 0 \end{vmatrix} = 2(0 - 24) - 3(0 - 6) + 5(16 - 1) = 45,$$

$$\text{the matrix of cofactors } \mathbf{C} = \begin{pmatrix} -24 & 6 & 15 \\ 20 & -5 & -5 \\ 13 & 8 & -10 \end{pmatrix}$$

$$\text{and the adjoint of } \mathbf{A}, \text{ i.e. } \mathbf{C}^T = \begin{pmatrix} -24 & 20 & 13 \\ 6 & -5 & 8 \\ 15 & -5 & -10 \end{pmatrix}$$

Then the inverse of $\mathbf{A}$ is given by

$$\mathbf{A}^{-1} = \begin{pmatrix} -\frac{24}{45} & \frac{20}{45} & \frac{13}{45} \\ \frac{6}{45} & -\frac{5}{45} & \frac{8}{45} \\ \frac{15}{45} & -\frac{5}{45} & -\frac{10}{45} \end{pmatrix} = \frac{1}{45} \begin{pmatrix} -24 & 20 & 13 \\ 6 & -5 & 8 \\ 15 & -5 & -10 \end{pmatrix}$$

Programme 5: Matrices

Inverse of a square matrix

Therefore, to form the inverse of a square matrix $\mathbf{A}$:

- (a) Evaluate the determinant of $\mathbf{A}$, i.e. $|\mathbf{A}|$
- (b) Form a matrix $\mathbf{C}$ of the cofactors of the elements of $|\mathbf{A}|$
- (c) Write the transpose of $\mathbf{C}$, i.e. $\mathbf{C}^T$, to obtain the adjoint of $\mathbf{A}$
- (d) Divide each element of $\mathbf{C}^T$ by $|\mathbf{A}|$
- (e) The resulting matrix is the inverse $\mathbf{A}^{-1}$ of the original matrix $\mathbf{A}$.

Inverse of a square matrix

Example:

Determine the inverse of the matrix $\mathbf{A} = \begin{pmatrix} 2 & 7 & 4 \\ 3 & 1 & 6 \\ 5 & 0 & 8 \end{pmatrix}$

$$\mathbf{A}^{-1} = \frac{1}{38} \begin{pmatrix} 8 & -56 & 38 \\ 6 & -4 & 0 \\ -5 & 35 & -19 \end{pmatrix}$$

Programme 5: Matrices

Finding the inverse of a 2×2 matrix

Suppose A is a 2×2 matrix and $|A| \neq 0$. Then the inverse of A can be found. If

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

then recall $|A| = ad - bc$. Now A^{-1} is given by

$$A^{-1} = \frac{1}{|A|} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Programme 5: Matrices

Inverse of a square matrix

Product of a square matrix and its inverse

Square matrices may have inverses. When a matrix A has an inverse, we say it is invertible and denote its inverse by A^{-1} . The inverse matrix satisfies

$$AA^{-1} = A^{-1}A = I.$$

If A and B are invertible matrices, then $(AB)^{-1} = B^{-1}A^{-1}$.

Furthermore, if A is invertible then so is A^T

$$\text{and } (A^T)^{-1} = (A^{-1})^T.$$

Programme 5: Matrices

Inverse of a 2x2 matrix

Examples:

1. Find the inverses of the matrices $\begin{pmatrix} 5 & 6 \\ 4 & 5 \end{pmatrix}$ and $\begin{pmatrix} 6 & 4 \\ 3 & 3 \end{pmatrix}$.
2. Prove that if A and B are same-sized invertible matrices, then $(AB)^{-1} = B^{-1}A^{-1}$.

Test exercise 5



The questions are all straightforward and based on the work covered. You will have no trouble.



1 If $\mathbf{A} = \begin{pmatrix} 2 & 4 & 6 & 3 \\ 1 & 7 & 0 & 4 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 3 & 5 & 2 & 7 \\ 9 & 1 & 6 & 3 \end{pmatrix}$ determine

(a) $\mathbf{A} + \mathbf{B}$ and (b) $\mathbf{A} - \mathbf{B}$.

2 Given that $\mathbf{A} = \begin{pmatrix} 6 & 0 & 4 \\ 1 & 5 & -3 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 & 9 \\ 8 & 0 \\ -4 & 7 \end{pmatrix}$ determine

(a) $3\mathbf{A}$, (b) $\mathbf{A}\mathbf{B}$, (c) $\mathbf{B}\mathbf{A}$.



3 If $\mathbf{A} = \begin{pmatrix} 2 & 3 & 5 \\ 1 & 7 & 4 \\ 8 & 0 & 6 \end{pmatrix}$, form the transpose $\mathbf{A}^T$ and determine the matrix product $\mathbf{A}^T\mathbf{I}$.

4 Show that the square matrix $\mathbf{A} = \begin{pmatrix} 3 & 2 & 4 \\ 1 & 5 & 3 \\ -1 & 8 & 2 \end{pmatrix}$ is a singular matrix.



5 If $\mathbf{A} = \begin{pmatrix} 1 & 4 & 3 \\ 6 & 2 & 5 \\ 1 & 7 & 0 \end{pmatrix}$, determine (a) $|\mathbf{A}|$ and (b) $\text{adj } \mathbf{A}$.

6 Find the inverse of the matrix $\mathbf{A} = \begin{pmatrix} 2 & 1 & 4 \\ 3 & 5 & 1 \\ 2 & 0 & 6 \end{pmatrix}$

Solutions on page 1138, Stroud

Review summary



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- 1 *Matrix* – a rectangular array of numbers (elements). [1]
- 2 *Order* – a matrix order of $(m \times n)$ denotes m rows and n columns. [1]
- 3 *Row matrix* – one row only. [2]
- 4 *Column matrix* – one column only. [2]
- 5 *Double suffix notation* – a_{34} denotes element in 3rd row and 4th column. [4]
- 6 *Equal matrices* – corresponding elements equal. [8]
- 7 *Addition and subtraction of matrices* – add or subtract corresponding elements. Therefore, for addition or subtraction, matrices must be of the same order. [10]
- 8 *Multiplication of matrices* [12]
 - (a) *Scalar multiplier* – every element multiplied by the same constant, i.e. $k[a_{ij}] = [ka_{ij}]$.
 - (b) *Matrix multiplier* – product **A.B** possible only if the number of columns in **A** equals the number of rows in **B**.

$$\overrightarrow{\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}} \cdot \begin{pmatrix} g & j \\ h & k \\ i & l \end{pmatrix} = \begin{pmatrix} ag + bh + ci & aj + bk + cl \\ dg + eh + fi & dj + ek + fl \end{pmatrix}$$

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Worked examples and exercises are in the text



- 9** *Transpose of a matrix* – rows and columns interchanged. Transpose of $\mathbf{A}$ is $\mathbf{A}^T$. [19]
- 10** *Square matrix* – of order $(m \times m)$ [21]
- (a) *Symmetric* if $a_{ij} = a_{ji}$, e.g. $\begin{pmatrix} 3 & 2 & 4 \\ 2 & 6 & 1 \\ 4 & 1 & 5 \end{pmatrix}$
- (b) *Skew symmetric* if $a_{ij} = -a_{ji}$, e.g. $\begin{pmatrix} 0 & 2 & 4 \\ -2 & 0 & 1 \\ -4 & -1 & 0 \end{pmatrix}$
- 11** *Diagonal matrix* – all elements zero except those on the leading diagonal. [21]
- 12** *Unit matrix* – a diagonal matrix with elements on the leading diagonal all unity, i.e. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ denoted by $\mathbf{I}$. [21]
- 13** *Null matrix* – all elements zero. [22]
- 14** *Cofactors* – minors of the elements of $|\mathbf{A}|$ together with the respective ‘place signs’ of the elements. [25]

Programme 5: Matrices

15 *Adjoint of a square matrix $\mathbf{A}$ – form matrix $\mathbf{C}$ of the cofactors of the elements of $|\mathbf{A}|$, then the adjoint of $\mathbf{A}$ is $\mathbf{C}^T$, i.e. the transpose of $\mathbf{C}$. $\therefore \text{adj } \mathbf{A} = \mathbf{C}^T$. [26]*

16 *Inverse of a square matrix $\mathbf{A}$ [28]*

$$\mathbf{A}^{-1} = \frac{\text{adj } \mathbf{A}}{|\mathbf{A}|} = \frac{\mathbf{C}^T}{|\mathbf{A}|}$$

17 *Product of a square matrix and its inverse [33]*

$$\mathbf{A}.\mathbf{A}^{-1} = \mathbf{A}^{-1}.\mathbf{A} = \mathbf{I}$$